

- **Non-modular:** Cheapest option for basic builds. Suitable for budget systems where cable clutter is not a concern. Examples include Antec Atom and Seasonic S12III.

## Other critical considerations

### Form factor and compatibility

Ensure your PSU fits in your case and matches the ATX, SFX or TFX standards. Always measure the case's PSU compartment and check the unit's length. For mini-ITX builds, SFX or SFX-L units from Corsair or SilverStone are a must. Some high-wattage ATX PSUs exceed 200 mm in length, which can obstruct drive cages or front-mounted radiators.



### Connector count and cable length

Verify that the PSU includes all necessary connectors: enough PCIe cables for multiple GPUs, SATA connectors for drives and Molex connectors for legacy devices. Cable length matters for large cases; look for extended cables or buy cable extensions for full-tower builds.

### Safety and reliability

Choose PSUs with robust protection features (SCP, OVP, OPP/OCP). Higher-end models may also include over-temperature protection (OTP) and inrush current protection. Good capacitors (often advertised as Japanese capacitors) and long warranties – ten years or more – indicate quality. Seasonic's PRIME Series offers a 12-year warranty. Corsair, Cooler Master and Thermaltake also provide long warranties on their premium lines.

### Noise and cooling

The PSU's cooling solution affects noise. Larger 135 mm or 140 mm fans typically spin slower and are quieter than 120 mm fans. Semi-passive or "hybrid" modes keep the fan off at low loads. Cybenetics LAMBDA ratings provide a standardized noise measurement. If silence is a priority, look for units rated A or A++ ( $\leq 25$  dB).

### Future-proofing and ATX 3.1

PCI Express 5.0 graphics cards and ATX 3.1 PSUs use new 12V-2×6 connectors capable of delivering up to 600 W through a single cable. Ensure that your PSU either ships with this connector or includes an adapter. Future-proofing also means choosing a PSU with additional wattage capacity (20–30 % above your current needs) and modular cabling for easy upgrades.

## Troubleshooting and maintenance tips

- **Regularly inspect cables and connectors.** Over time, connectors can loosen due to vibration. Re-seat cables during annual clean-ups.
- **Clean dust filters and vents.** Dust accumulation impedes airflow and can overheat the PSU. Vacuum or blow out dust filters every few months.
- **Monitor voltages and temperatures.** Use motherboard BIOS monitoring or software tools to ensure voltages remain within  $\pm 5\%$  of nominal values. Unstable voltages may indicate a failing PSU.
- **Plan for upgrades.** If you add a new GPU or CPU, revisit your wattage calculations and ensure the PSU provides enough headroom.
- **Be aware of warranty terms.** Keep the purchase receipt and register your PSU if required. Many brands offer 5- to 12-year warranties.



A power supply might not be the flashiest component in your 2025 PC build, but it is arguably the most important. Choosing wisely involves understanding wattage requirements, efficiency certifications, and modularity options. Use available calculators to size your PSU, aim for 80 PLUS Gold or higher for balanced efficiency, and invest in reputable brands that offer robust protection and long warranties. A quality PSU not only delivers stable power to today's components but also leaves room to grow, ensuring your system stays reliable through years of gaming, creating and everyday computing. 